

MEng Computer and Electronic engineering student at Loughborough, seeking a 2027 industrial placement in embedded systems or FPGAs

Employment and Work Experience

Technical theatre: Trinity school

October 2018 – July 2025

- Developed backstage production skills for 3 shows a year (including 4 musicals), running sound design and operation for 5 years and a further year on projections.

Website designer

June 2025 – July 2025

- Designed and built a functioning website using HTML, CSS and JS with several pages and easy navigation through a responsive navigation bar.

Hockey coach at Spencer Lynx hockey club

September 2023 – July 2025

- Coached hockey for 9 – 11-year-olds every week.

Activity Instructor at Activ Camps

April 2024 – December 2024

- Worked in a team as an activity instructor helping run activities for 4 – 17-year-olds trying to keep them engaged and help them enjoy the activity while keeping them safe.

InvestIN engineering course: UCL university campus

August 2024

- Completed a two-week course where I networked with many professional engineers, discussing the economic and environmental impacts of engineering, while studying many of the different sectors and winning the final competition with an app concept where you could try different hobbies.

Video Editor

June 2022 – August 2022

- Designed and edited a 2-hour montage video for a 20th wedding anniversary.
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Projects

Datum Engineering Notebook

July 2026 – Present

- iPad engineering notebook built using Swift for capturing sketches, schematics and STL models alongside parts and budget tracking.

Predictive bus routing

March 2026

- Grand Finalist (top 36) in an international competition, the Engineers Without Borders Design Challenge. We built an XGBoost model to predict bus stop demand and a CVRP routing optimiser to cut wait times, with a custom-built LED map display to show the results running on a DE1-SoC FPGA coded in Verilog.

Vaultly

February 2026 – Present

- Built and self-hosted a full stack document vault on a home server with CI/CD, currently beta testing at getvaultly.uk

IMC Prosperity 4

April 2026 – May 2026

- Ranked 700th of ~18,900 (top 3%) in a global algorithmic trading competition; Helped design market making, statistical arbitrage and mean reversion strategies across multiple rounds in a two-person team.

Machine Learning – to unwrap ancient scrolls

February 2026

- Collaborated on designing and training a machine learning algorithm to unwrap ancient scrolls that cannot be unwrapped physically as part of a Kaggle competition.

Data collection Terminal

October 2025

- Designed and coded it in C for an Arduino system that has connected sensors and an updating display, connected to an ESP32 that transmits this data to an online website to read the data in real time from anywhere.

Lightsaber

June 2025

- Engineered a fully functional combat lightsaber, designing the handle on professional workshop tools and making the blade using LEDs and soldering all the electronics to an Arduino microcontroller to code animations for turning the blade on and off using C.

Tic Tac Toe – in Minecraft

April 2024

- Used Minecraft as a 3-dimensional way to emulate an embedded system using custom built architecture based on real world Boolean logic that created a playable version of Tic Tac Toe and have since extended the architecture to connect 4 and pong
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Education

University: Loughborough University

September 2025 – June 2030

First year Computer and Electronic Engineer (June 2026):

- First year – *First Class*

Secondary school: Trinity School Croydon

September 2018 – June 2025

A-Levels (June 2025):

- Computer Science, Design and Technology, Maths – *B, A, C*

GCSEs (June 2023):

- Maths, Physics, Chemistry, Biology, Design and Technology, Computer Science, Geography, English Literature, English Language, Spanish – *9, 8, 8, 8, 8, 8, 7, 6, 6, 5*
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Key skills

- **Hardware:** Arduino, DE1-Soc FPGA, Verilog, Raspberry Pi, ESP-32.
 - **Programming:** C#, C, python, Java, HTML, CSS, C++, JavaScript.
 - **Video editing:** Da Vinci resolve, Filmora, Final Cut, Blender (3D VFX).
 - **Engineering applications:** SolidWorks, GitHub, NI Multisim, FEMM, RS DesignSpark.
 - **Professional Applications:** Word, Power Point, Excel, Outlook, OneNote.
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Interests

Mountain Biking – I tackle technical trails and long-distance routes such as the London to Brighton.

Magic – I practice close-up magic, developing strong presentation, confidence, and audience engagement skills.

Hiking & Camping – I enjoy multi-day hiking and camping trips, strengthening my planning, navigation, and self-sufficiency skills and have completed my Bronze, Silver and Gold Duke of Edinburgh award.

Scuba Diving – I am a certified scuba diver with experience diving in varied conditions across several countries.

Hockey – I play hockey every week, contributing to team performance through communication, teamwork, and strategic thinking.

Guitar – I play guitar in my spare time, continually improving through self-directed learning and practice.

Solving Puzzles (e.g. Sudoku) – I regularly solve logic puzzles such as Sudoku and the LinkedIn daily puzzles to sharpen problem-solving and analytical thinking skills.
